

## 1) Create Table

```
import sqlite3

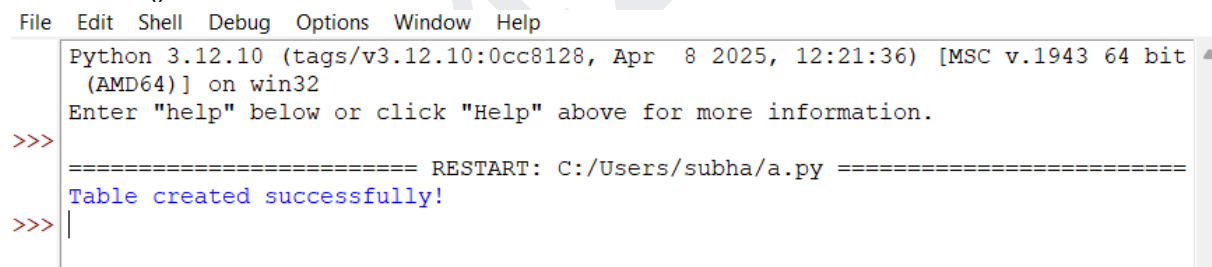
# Connect to SQLite database (creates file if it doesn't exist)
conn = sqlite3.connect("student.db")

# Create cursor
cur = conn.cursor()

# Create Student table
cur.execute("""
CREATE TABLE IF NOT EXISTS Student(
    id INTEGER PRIMARY KEY,
    name TEXT NOT NULL,
    age INTEGER,
    department TEXT
)
""")

print("Table created successfully!")

conn.commit()
conn.close()
```



The screenshot shows a Python IDE window titled "Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32". The menu bar includes File, Edit, Shell, Debug, Options, Window, and Help. The main text area displays the output of a script execution. It starts with a prompt "Enter 'help' below or click 'Help' above for more information." followed by a separator line "===== RESTART: C:/Users/subha/a.py =====". Below this, the text "Table created successfully!" is shown in blue, indicating a successful execution. The prompt ">>>" is visible at the end of the line.

```
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/subha/a.py =====
Table created successfully!
>>>
```

## 2) Insert Table

```
import sqlite3

conn = sqlite3.connect("student.db")
cur = conn.cursor()

cur.execute("""
INSERT INTO Student(id, name, age, department)
VALUES (1, 'Ravi', 20, 'CSE')
""")

conn.commit()
conn.close()

print("One record inserted!")
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/insert.py =====
One record inserted!
|
```

## 3) Multiple Records

```
import sqlite3

conn = sqlite3.connect("student.db")
cur = conn.cursor()

students = [
    (2, 'Priya', 21, 'IT'),
    (3, 'Amit', 19, 'ECE'),
    (4, 'Neha', 22, 'CSE')
]

cur.executemany("""
INSERT INTO Student(id, name, age, department)
VALUES (?, ?, ?, ?)
""", students)

conn.commit()
print("Multiple records inserted!")

conn.close()
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/multiple.py =====  
Multiple records inserted!
```

#### 4)Display all records

```
import sqlite3  
  
conn = sqlite3.connect("student.db")  
cur = conn.cursor()  
  
cur.execute("SELECT * FROM Student")  
  
rows = cur.fetchall()  
  
for row in rows:  
    print(row)  
  
conn.close()
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/display.py =====  
(1, 'Ravi', 20, 'CSE')  
(2, 'Priya', 21, 'IT')  
(3, 'Amit', 19, 'ECE')  
(4, 'Neha', 22, 'CSE')  
>>> |
```

#### 5)Display Name and Age where Age > 20

```
import sqlite3  
  
conn = sqlite3.connect("student.db")  
cur = conn.cursor()  
  
cur.execute("""  
SELECT name, age  
FROM Student  
WHERE age > 20  
""")
```

```
rows = cur.fetchall()
```

```
for row in rows:  
    print(row)
```

```
conn.close()
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/age.py =====  
('Priya', 21)  
('Neha', 22)  
>>>
```

## 6) Update query

```
import sqlite3
```

```
conn = sqlite3.connect("student.db")  
cur = conn.cursor()
```

```
cur.execute("""  
UPDATE Student  
SET age = 22  
WHERE id = 1  
""")
```

```
conn.commit()
```

```
print("Record updated!")
```

```
conn.close()
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/update.py =====  
Record updated!
```

## 7)DELETE Query

Delete record with ID = 3

```
import sqlite3

conn = sqlite3.connect("student.db")
cur = conn.cursor()

cur.execute("""
DELETE FROM Student
WHERE id = 3
""")

conn.commit()

print("Record deleted!")

conn.close()
```

```
>>> ===== RESTART: C:/Users/subha/Desktop/Query processing/delete.py =====
Record deleted!
>>>
```

## 8) View Final Table

```
import sqlite3

conn = sqlite3.connect("student.db")
cur = conn.cursor()

cur.execute("SELECT * FROM Student")

rows = cur.fetchall()

print("Final Table")
for row in rows:
    print(row)

conn.close()
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/view.py =====
Final Table
(1, 'Ravi', 22, 'CSE')
(2, 'Priya', 21, 'IT')
(4, 'Neha', 22, 'CSE')
>>> |
```

## APPLICATION

Student Management System

Student ID

Name

Age

Department

Add

Update

Delete

Clear

View Students

| ID  | Name   | Age | Department |
|-----|--------|-----|------------|
| 1   | Ravi   | 22  | CSE        |
| 2   | Priya  | 21  | IT         |
| 4   | Neha   | 22  | CSE        |
| 145 | subhaa | 19  | AI&DS      |